

# Blind Speech-Enhancement Floors under Harmonic Rotor Noise

A five-architecture, two-pass baseline study on drone and harmonic noise

Dmitrii Mukhutdinov<sup>a</sup>

<sup>a</sup>Harmonic Noise Suppression Project

## Abstract

We establish the **blind** (no rotor-speed side information) speech-enhancement floor on our ultra-low-SNR (−30...0 dB) harmonic-noise data with five modern architectures, trained in two passes — on drone noise only, and on a category-uniform mix of nine harmonic-noise families — and scored on a fixed, held-out validation set against noisy-input and Wiener anchors. The floor is strongly **architecture-bound**: MP-SENet (parallel magnitude+phase) is the strongest baseline, and both ported models (MP-SENet, TF-GridNet) beat the in-house Edge-BS-RoFormer — even while compute-limited — while the classic complex-UNet (DCUNet) only removes noise energy without restoring intelligibility. Whether training on **diverse** harmonic noise helps on drone noise is **capacity-dependent** — it aids only the capacity-limited DCUNet and dilutes all three stronger ports (MP-SENet, TF-GridNet, Edge-BS-RoFormer). These floors gate later rotor-informed claims.

**Keywords:** speech enhancement; drone ego-noise; ultra-low SNR; SI-SDR; blind baselines.

## Contents

|                                   |   |
|-----------------------------------|---|
| Introduction .....                | 3 |
| Methods .....                     | 3 |
| Data and mixing .....             | 3 |
| Architectures .....               | 3 |
| Loss and training .....           | 3 |
| Evaluation .....                  | 3 |
| Results .....                     | 4 |
| Per-category harmonic floor ..... | 5 |
| Discussion .....                  | 6 |
| Status and caveats .....          | 6 |
| Bibliography .....                | 7 |

## Introduction

Speech enhancement under the harmonic ego-noise of rotating sources (drone motors and propellers) is an extreme low-SNR problem: at onboard microphones the target speech sits  $-30$  to  $0$  dB below dense, structured rotor noise. Before any rotor-speed-informed method can claim a benefit, we need an honest **blind** floor — how well modern architectures do with no side information — measured on our own data with strong anchors. This report establishes that floor.

We ask two questions. **(i)** Which modern architectures set the blind floor, and by how much do they beat trivial anchors (noisy input, Wiener)? **(ii)** Does training on a **diverse** set of harmonic noises help on drone noise (transferable harmonic structure) or hurt (capacity dilution)? We answer both with a five-architecture set trained in two passes and evaluated per input SNR.

## Methods

### Data and mixing

Speech is LibriSpeech train-clean-100 (25 speakers held out for validation). Noise is drawn online each step and mixed at  $\text{SNR} \sim \mathcal{U}(-30, 0)$  dB with random\_gain/random\_polarity augmentation, 16 kHz mono, 1 s chunks. Two noise regimes define the passes:

- **Pass A (drone only):** DREGON + Michael’s real drone recordings plus open drone-noise datasets, roughly uniform over sub-datasets.
- **Pass B (all harmonic):** category-uniform over nine families — drone, MIMII (industrial), MIMII-DG, aircraft (AeroSonicDB), motors (HUSTmotor, KAIST), horns (HornBase) — so the 258 GiB MIMII cannot dominate.

The noise pool streams lazily from object storage at shard granularity, capped at 24 shards per dataset (a bounded, diverse subset — an uncapped random-shard-per-sample draw would stream the whole 258 GiB MIMII over a run). Both passes are scored on the **same** fixed, published validation set: held-out noise recordings (recording-level split)  $\times$  held-out speakers, SNR grid  $\{-30, -25, -20, -15, -10, -5, 0\}$  dB, 50 mixtures/point, deterministic. Every table carries **noisy-input** and **Wiener** (`scipy.signal.wiener`) anchors.

### Architectures

Five blind (no-RPS) waveform-in/waveform-out models: **Edge-BS-RoFormer** (band-split rotary transformer, our Paper-1 model), **TF-GridNet** (dense full+sub-band dual-path, mid-size  $\sim 8.4$  M, ported), **MP-SENet** (parallel magnitude+phase,  $\sim 1.7$  M, ported), **DCUNet** (complex U-Net, the 2023 benchmark winner [1] — a continuity anchor), and **SGMSE+** (score-based diffusion, ported, trained from scratch; deferred; see the caveats below).

### Loss and training

An initial masked-MSE pass produced floors **at** the noisy level (at ultra-low SNR, MSE rewards attenuation-toward-silence, improving SDR but not intelligibility). We therefore train with a metric-aligned composite: negative SI-SDR + a multi-resolution STFT term. Training length matches the 2023 benchmark [1]: early-stop patience  $N_E = 30$ , LR-plateau patience  $N_\alpha = 15$ ,  $\sim 1300$  steps/epoch, cap 150 epochs, AdamW at  $10^{-3}$ , amp off (complex STFT). Jobs run on A100 GPUs.

### Evaluation

The single evaluation path scores each checkpoint per SNR with SI-SDR, SDR, PESQ (wideband) and extended STOI. For a **fair** cross-model comparison all models are scored on the same balanced 25-mixtures/SNR subset (SI-SDR is high-variance at 0 dB, so mixing 25- and 50-clip means across models is unfair; the 25-subset ranking is stable, absolute values at 0 dB carry  $\pm$  a few dB).

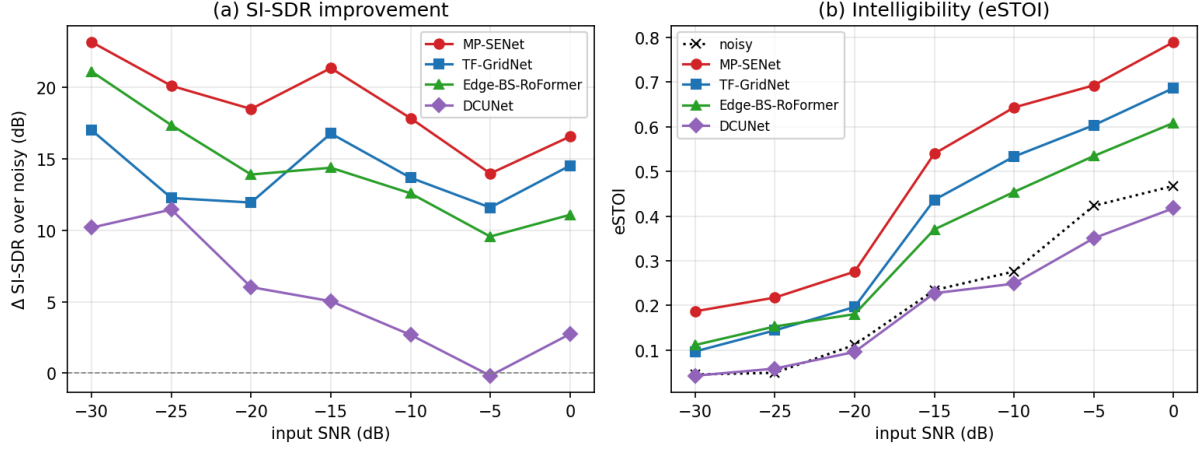


Figure 1. Blind SE floor on SE-valid-drone (Pass A). (a) SI-SDR improvement over the noisy input vs input SNR. (b) Absolute eSTOI (intelligibility); the dotted line is the noisy input. MP-SENet and TF-GridNet — both ported — lead the in-house Edge-BS-RoFormer; DCUNet tracks the noisy eSTOI.

Table 1. SI-SDR improvement over the noisy input (dB), per input SNR, uniform 25-mixtures/point. Bold = best. Every model clears the Wiener anchor at  $\geq -10$  dB.

| SNR | Wiener | <b>MP-SENet</b> | TF-GridNet | Edge-BS-RoF | DCUNet-A | DCUNet-B |
|-----|--------|-----------------|------------|-------------|----------|----------|
| -30 | -0.1   | <b>+23.2</b>    | +17.0      | +21.1       | +10.2    | +11.9    |
| -20 | +0.1   | <b>+18.5</b>    | +11.9      | +13.9       | +6.0     | +6.0     |
| -15 | +0.1   | <b>+21.4</b>    | +16.8      | +14.4       | +5.0     | +7.6     |
| -10 | +0.1   | <b>+17.9</b>    | +13.7      | +12.6       | +2.7     | +3.9     |
| -5  | +0.1   | <b>+14.0</b>    | +11.6      | +9.6        | -0.2     | +0.7     |
| 0   | +0.1   | <b>+16.6</b>    | +14.5      | +11.1       | +2.7     | +2.2     |

Table 2. Absolute eSTOI (higher = more intelligible), noisy input vs models.

| SNR | noisy | <b>MP-SENet</b> | TF-GridNet | Edge-BS-RoF | DCUNet |
|-----|-------|-----------------|------------|-------------|--------|
| -15 | 0.234 | <b>0.540</b>    | 0.436      | 0.370       | 0.228  |
| -10 | 0.276 | <b>0.643</b>    | 0.533      | 0.454       | 0.249  |
| -5  | 0.423 | <b>0.693</b>    | 0.603      | 0.534       | 0.350  |
| 0   | 0.468 | <b>0.789</b>    | 0.687      | 0.609       | 0.418  |

## Results

Figure 1 and Table 1 summarise Pass A on the drone validation set. The spread across architectures is large and the ranking is unambiguous: **MP-SENet** > **TF-GridNet**  $\gtrsim$  **Edge-BS-RoFormer**  $\gg$  **DCUNet**.

**Intelligibility (eSTOI).** The models separate most clearly on eSTOI (Table 2). MP-SENet nearly doubles noisy intelligibility at -15 dB (0.234  $\rightarrow$  0.540) and reaches 0.789 at 0 dB; TF-GridNet and Edge-BS-RoFormer follow; DCUNet stays at the noisy level — it removes noise energy (SI-SDR/PESQ rise) without restoring speech.

**Diversity verdict — capacity-dependent (Table 3).** We compare each model’s drone-only pool (Pass A) against the all-harmonic pool (Pass B), scored on the drone valid (both passes select the best drone-valid checkpoint, and both are the same compute-limited regime, so the delta is a fair within-

Table 3. Diversity delta:  $\Delta$  SI-SDR of Pass B (all-harmonic) minus Pass A (drone-only) on the drone valid (dB). Positive (bold) = diversity helps. It helps only the capacity-limited DCUNet; the three stronger ports are hurt or unmoved.

| model (B–A)   | –30         | –25  | –20  | –15         | –10         | –5          | 0    |
|---------------|-------------|------|------|-------------|-------------|-------------|------|
| DCUNet (weak) | <b>+1.7</b> | –0.0 | –0.0 | <b>+2.5</b> | <b>+1.2</b> | <b>+0.9</b> | –0.6 |
| MP-SENet      | –3.6        | –4.6 | –4.3 | –2.6        | –1.4        | –0.1        | +0.0 |
| TF-GridNet    | –1.1        | –0.8 | –1.3 | –2.2        | –3.6        | –2.3        | –1.3 |
| Edge-BS-RoF   | –3.6        | –4.5 | –6.3 | –4.4        | –2.9        | –3.0        | –1.3 |

Table 4. Per-family blind floor on SE-valid-harmonic (Pass B): SI-SDR improvement over noisy (dB), mean over the SNR grid, sorted easiest→hardest. Harmonic/tonal families (motors, aircraft, horns, drone) are recovered far better than stochastic industrial noise (MIMII).

| family   | DCUNet | <b>MP-SENet</b> | TF-GridNet  | Edge-BS-RoF |
|----------|--------|-----------------|-------------|-------------|
| motors   | +3.9   | <b>+19.6</b>    | +16.9       | +12.1       |
| aircraft | +10.4  | <b>+16.6</b>    | +14.4       | +16.5       |
| drone    | +5.9   | <b>+16.5</b>    | +12.2       | +10.2       |
| horns    | +2.5   | <b>+15.7</b>    | +11.4       | +9.1        |
| mimii    | +1.2   | <b>+9.9</b>     | +8.1        | –0.9        |
| mimii-dg | +1.2   | +4.7            | <b>+6.9</b> | +5.3        |

architecture comparison). The verdict splits cleanly by capacity: for the weak **DCUNet**, Pass B  $\geq$  Pass A at most SNRs ( $\Delta$  SI-SDR at –15 dB: +2.5) — extra harmonic data helps a model that under-fits drone noise alone. But **all three stronger ports are hurt or unmoved**: Edge-BS-RoFormer and TF-GridNet lose SI-SDR at every SNR and eSTOI across the board (Edge-BS-RoF eSTOI –0.036 at –15 dB, TF-GridNet –0.031), while MP-SENet loses SI-SDR at low SNR and is flat on eSTOI. Under an equal training budget, diverse noise dilutes drone-specific capacity; the transferable structure of rotating-source harmonics helps only when the model is capacity-limited on the target. A capable model does better **focusing** on the target noise.

**Loss and length matter.** On DCUNet, masked-MSE gave a floor essentially at the noisy level; the SI-SDR+MRSTFT composite with the paper-matched schedule lifted the –15 dB SI-SDR gain from +1.6 dB (an early-stopping-truncated 12-minute run) to +5.0 dB — a reminder that a “baseline” is a claim about training as much as architecture.

### Per-category harmonic floor

Evaluating the Pass-B (all-harmonic) models per noise family on SE-valid-harmonic (Table 4, mean over the SNR grid) shows a clean, model-independent difficulty ordering: the **more harmonic** the source, the more enhancement recovers. The strongly-tonal families — motors, aircraft, horns — are the easiest (MP-SENet lifts SI-SDR +15.7...+19.6 dB, eSTOI to 0.52...0.57), drone sits just below, and the **stochastic** industrial machine noise of MIMII / MIMII-DG is hardest (MP-SENet eSTOI only 0.29...0.34). This is direct evidence for the project’s premise: it is harmonic structure — not loudness — that a blind model can exploit, so a rotating-source target is a favourable case, and the broadband-machine families bound the low end.

The Pass B – Pass A transfer per family (Table 5) echoes the capacity story from the drone valid, now resolved by family. Adding the noisy, **broadband** MIMII families to a fixed budget mostly **hurts** the strong models even on those very families (Edge-BS-RoFormer –11.9 dB on MIMII, MP-

Table 5. Per-family transfer:  $\Delta$  SI-SDR of Pass B minus Pass A (dB) on SE-valid-harmonic. Harmonic families (motors, horns) gain across all archs; stochastic MIMII families lose for the strong models.

| family (B-A) | DCUNet | MP-SENet | TF-GridNet | Edge-BS-RoF |
|--------------|--------|----------|------------|-------------|
| motors       | +6.9   | +2.0     | +6.9       | +2.7        |
| horns        | +0.6   | +3.8     | +3.9       | +1.3        |
| aircraft     | +4.9   | -4.3     | +1.9       | -0.0        |
| drone        | +0.3   | -1.8     | -1.8       | -3.7        |
| mimii-dg     | +1.3   | -7.6     | -0.2       | -4.3        |
| mimii        | -0.8   | -4.9     | -0.0       | -11.9       |

SENet -7.6 on MIMII-DG) — they cannot fit stochastic noise and lose ground focusing on it — whereas the **harmonic** motors and horns families benefit from in-domain exposure across every architecture (all four positive). The weak DCUNet, which under-fits throughout, gains almost everywhere. Diverse data helps when the added families share exploitable structure and the model has spare capacity; it hurts when neither holds.

## Discussion

The blind floor on our data is set by the **newer** speech-enhancement architectures: MP-SENet’s explicit magnitude-and-phase decoders are the strongest, and both ports beat the in-house Edge-BS-RoFormer. This is a useful recalibration — the strongest blind baseline is not the band-split transformer we had been treating as SOTA. The intelligibility gap between models that **restore** speech (MP-SENet, TF-GridNet, Edge-BS-RoFormer) and one that merely **denoises** (DCUNet) is the headline: at these SNRs, SI-SDR and PESQ can rise while eSTOI does not, so intelligibility metrics are the ones that separate real enhancement from attenuation.

That MP-SENet and TF-GridNet win **while compute-limited** (the caveats below) means their true ceiling is higher still. The diversity result cautions against a one-size-fits-all data recipe: broadening training to many harmonic-noise families helps only the capacity-limited DCUNet and **hurts or fails to help** all three stronger ports on the target noise. The split is clean across every model we ran, so data breadth should scale with — not substitute for — architecture and training budget.

## Status and caveats

The MP-SENet and TF-GridNet numbers are **lower bounds**: amp-off fp32 makes them  $\sim 10 \times$  slower per step than DCUNet (MP-SENet 1.5 it/s; TF-GridNet slower at batch 4), so both hit the wall-clock wall before convergence; their best checkpoints were recovered from object storage. fp16 training (STFT kept in fp32) would let them converge and would likely widen their lead. The Pass B deltas for the compute-limited MP-SENet and TF-GridNet are read from their best-drone-valid checkpoints while their runs continue, matched against the same-regime Pass A checkpoints; the direction of the diversity effect is stable, absolute magnitudes may tighten as they converge. SGMSE+ (65 M, score-matching) is deferred — it needs a bespoke training loop and a multi-day budget. The per-category harmonic tables use a balanced 15-clips/(family, SNR) subsample for CPU tractability; the difficulty ordering is stable, per-cell absolutes carry the usual small-sample variance.

## Bibliography

- [1] D. Mukhutdinov, A. Alex, L. Wang, and A. Cavallaro, “Deep Learning Models for Single-Channel Speech Enhancement on Drones,” *IEEE Access*, vol. 11, pp. 22993–23007, 2023, doi: 10.1109/ACCESS.2023.3253719.